Magnetic fields

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Interpret magnetic field patterns and compass direction.

SUCCESS CRITERIA - I can...

1. Label north and south poles.
2. Trace external field direction from north toward south.
3. Explain why field-line drawings are models.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A 5 Ω resistor is in series with two parallel 6 Ω resistors across 16 V. Find parallel equivalent, total resistance and supply current.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $B_{net,\,x} = B_{1,\,x} + B_{2,\,x}$; $B_{net,\,y} = B_{1,\,y} + B_{2,\,y}$. Add field components at the same point. A compass follows the net field. Outside a bar magnet, arrows run north to south.

Worked reasoning: Outside a bar magnet, field arrows run from the north pole toward the south pole. A compass's north-seeking end follows the local field direction.

Your turn: Draw curved external field lines and two small compass arrows at different points.

Marzano check - circle: ① ② ③

03 Represent and interpret

Label N and S on the bar magnet. Sketch external field arrows and the direction a nearby compass would point.



Marzano check - circle: ① ② ③

04 Solve and justify

A compass between two identical magnets is observed to have no stable preferred direction at one location. Explain what field superposition could mean there; name a limitation of that observation.

Marzano check - circle: ① ② ③

05 Analyze the reasoning

A student draws field lines that cross at one point. Explain why this is inconsistent with a single net field direction at that point.

Marzano check - circle: 1 2 3

06 Choose a method

In separate cases, an ideal blackbody temperature becomes 4 times as large and a sound frequency becomes 6 times as large at fixed sound speed. Compare the two wavelength changes.

Marzano check - circle: 1 2 3

07 Exit check

If a bar magnet is cut in half, do you obtain an isolated north pole and an isolated south pole? Explain the usual result.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Two magnets face N toward N. Sketch the field qualitatively and predict the force. Then reverse one magnet and explain both changes.

Marzano check - circle: 1 2 3